

ECON345 Homework 7: Loops and Debugging

Due: October 15th, 11:59 PM

Develop your code in an R script first if that helps you think, then answer the questions (including your code) in an R Markdown file. Comment your code to explain what you're doing — partial credit rewards a reasonable, well-documented attempt, not just a correct final answer. For any part you can't produce output for, briefly explain what you tried and what went wrong.

Question 1: Writing a Loop (15 points)

A professor has the exam scores for a class of 8 students:

```
exam_scores <- c(72, 88, 91, 65, 79, 100, 58, 83)
```

Question 1.1

Write a for loop that prints a letter grade for every score in `exam_scores`, using the usual cutoffs (90+ is A, 80-89 is B, 70-79 is C, 60-69 is D, below 60 is F).

Question 1.2

Modify your loop so that, instead of just printing each grade, it **saves** every grade into a new vector called `letter_grades`.

Question 1.3

Your loop in 1.2 almost certainly indexed `exam_scores` using something like `seq_along(exam_scores)` or `1:length(exam_scores)`. Why is `seq_along()` the safer choice? Describe a scenario where `1:length(x)` would cause a bug that `seq_along(x)` would not.

Question 2: Cumulative Loops (15 points)

A hiker recorded the distance (in miles) covered each day of a 7-day trip:

```
daily_miles <- c(8, 12, 6, 15, 9, 11, 7)
```

Question 2.1

Write a for loop that computes the hiker's **cumulative distance** after each day (i.e., the running total), saving every intermediate result in a vector called `cumulative_miles`.

Question 2.2

On which day did the hiker cross the halfway point of their total trip distance? Answer using code — don't just eyeball it.

Question 2.3

Suppose the loop in 2.1 was written without initializing `cumulative_miles <- NULL` before the loop. What would happen when you tried to run it? Why does the initialization step matter?

Question 3: Find the Bug (10 points)

A classmate wrote the following function, which should return `TRUE` if a number is even and `FALSE` otherwise. It doesn't work correctly for every input.

```
is_even <- function(x) {  
  remainder = x %% 2  
  if (remainder = 0) {  
    is_even_result <- TRUE  
  } else {  
    is_even_result <- FALSE  
  }  
}
```

Question 3.1

Which bug category from lecture does this fall into? Identify every bug in the function (there is more than one).

Question 3.2

Rewrite the function so it works correctly, and show that it gives the right answer for both an even and an odd number.

Question 4: Debugging Strategy (10 points)

Question 4.1

You run a 40-line script and get an error on the last line. Describe two concrete steps you'd take to find which earlier line actually caused the problem.

Question 4.2

What does `browser()` let you do that a `print()` statement doesn't? Give an example of a situation where you'd reach for `browser()` instead of adding more `print()` statements.

Submission

Upload two files to Canvas before the deadline:

- your `week_07.Rmd`, and

- the **knitted PDF**.

As always: comment your code, and if you get stuck, show your work. If your `.Rmd` won't knit, get each piece working before you move on to the next, and restart R and knit clean if results look stale.